

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_squared_error, r2_score

from sklearn.ensemble import RandomForestRegressor
```

```
# Load dataset
df = pd.read_csv("Dataset .csv")

# Display first 5 rows
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Currency	Has Table booking	Has Online delivery	Is delivering now	Switch to order menu	Price range	Aggregate rating	Rating color	Rating text	Votes
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botswana Pula(P)	Yes	No	No	No	3	4.8	Dark Green	Excellent	314
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botswana Pula(P)	Yes	No	No	No	3	4.5	Dark Green	Excellent	591
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botswana Pula(P)	Yes	No	No	No	4	4.4	Green	Very Good	270
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botswana Pula(P)	No	No	No	No	4	4.9	Dark Green	Excellent	365
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botswana Pula(P)	Yes	No	No	No	4	4.8	Dark Green	Excellent	229

5 rows × 21 columns

```
# Dataset shape
print("Shape:", df.shape)

# Column names
print("\nColumns:\n")
print(df.columns)
```

```
Shape: (9551, 21)

Columns:

Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
      'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
      'Average Cost for two', 'Currency', 'Has Table booking',
      'Has Online delivery', 'Is delivering now', 'Switch to order menu',
      'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
      'Votes'],
      dtype='object')
```

```
# Select useful columns

data = df[[
    'Restaurant Name',
    'City',
    'Cuisines',
    'Average Cost for two',
    'Price range',
    'Votes',
    'Aggregate rating'
]]

data.head()
```

	Restaurant Name	City	Cuisines	Average Cost for two	Price range	Votes	Aggregate rating
0	Le Petit Souffle	Makati City	French, Japanese, Desserts	1100	3	314	4.8
1	Izakaya Kikufuji	Makati City	Japanese	1200	3	591	4.5
2	Heat - Edsa Shangri-La	Mandaluyong City	Seafood, Asian, Filipino, Indian	4000	4	270	4.4
3	Ooma	Mandaluyong City	Japanese, Sushi	1500	4	365	4.9
4	Sambo Kojin	Mandaluyong City	Japanese, Korean	1500	4	229	4.8

Next steps:

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[New interactive sheet](#)

```
# Check missing values
data.isnull().sum()
```

	0
Restaurant Name	0
City	0
Cuisines	9
Average Cost for two	0
Price range	0
Votes	0
Aggregate rating	0

dtype: int64

```
# Fill missing categorical values

data['Cuisines'] = data['Cuisines'].fillna('Unknown')
data['City'] = data['City'].fillna('Unknown')

# Check again
data.isnull().sum()
```

```
/tmp/ipykernel_5479/1162277834.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Cuisines'] = data['Cuisines'].fillna('Unknown')
/tmp/ipykernel_5479/1162277834.py:4: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['City'] = data['City'].fillna('Unknown')
```

	0
Restaurant Name	0
City	0
Cuisines	0
Average Cost for two	0
Price range	0
Votes	0
Aggregate rating	0

dtype: int64

```
# Create label encoder
encoder = LabelEncoder()

# Encode categorical columns
data['Restaurant Name'] = encoder.fit_transform(data['Restaurant Name'])

data['City'] = encoder.fit_transform(data['City'])

data['Cuisines'] = encoder.fit_transform(data['Cuisines'])

data.head()
```

```
/tmp/ipykernel_5479/501042064.py:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Restaurant Name'] = encoder.fit_transform(data['Restaurant Name'])
/tmp/ipykernel_5479/501042064.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['City'] = encoder.fit_transform(data['City'])
/tmp/ipykernel_5479/501042064.py:9: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
data['Cuisines'] = encoder.fit_transform(data['Cuisines'])
```

	Restaurant Name	City	Cuisines	Average Cost for two	Price range	Votes	Aggregate rating	
0		3748	73	920	1100	3	314	4.8
1		3172	73	1111	1200	3	591	4.5
2		2896	75	1671	4000	4	270	4.4
3		4707	75	1126	1500	4	365	4.9
4		5523	75	1122	1500	4	229	4.8

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```
# Features
X = data.drop('Aggregate rating', axis=1)

# Target
y = data['Aggregate rating']
```

```
# Split dataset

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("Training data:", X_train.shape)
print("Testing data:", X_test.shape)

Training data: (7640, 6)
Testing data: (1911, 6)
```

```
# Create model
model = RandomForestRegressor()

# Train model
model.fit(X_train, y_train)
```

RandomForestRegressor

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RandomForestRegressor()

```
# Predict ratings
y_pred = model.predict(X_test)

# Display predictions
print(y_pred[:10])

[2.925 3.052 3.192 3.664 3.    0.    3.112 0.    2.937 3.565]
```

```
# Mean Squared Error
mse = mean_squared_error(y_test, y_pred)

# R2 Score
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

Mean Squared Error: 0.098458802823945
R2 Score: 0.9567424920146036

```
# Feature importance

importance = model.feature_importances_

features = X.columns

importance_df = pd.DataFrame({
    'Feature': features,
    'Importance': importance
})

importance_df = importance_df.sort_values(
    by='Importance',
    ascending=False
)

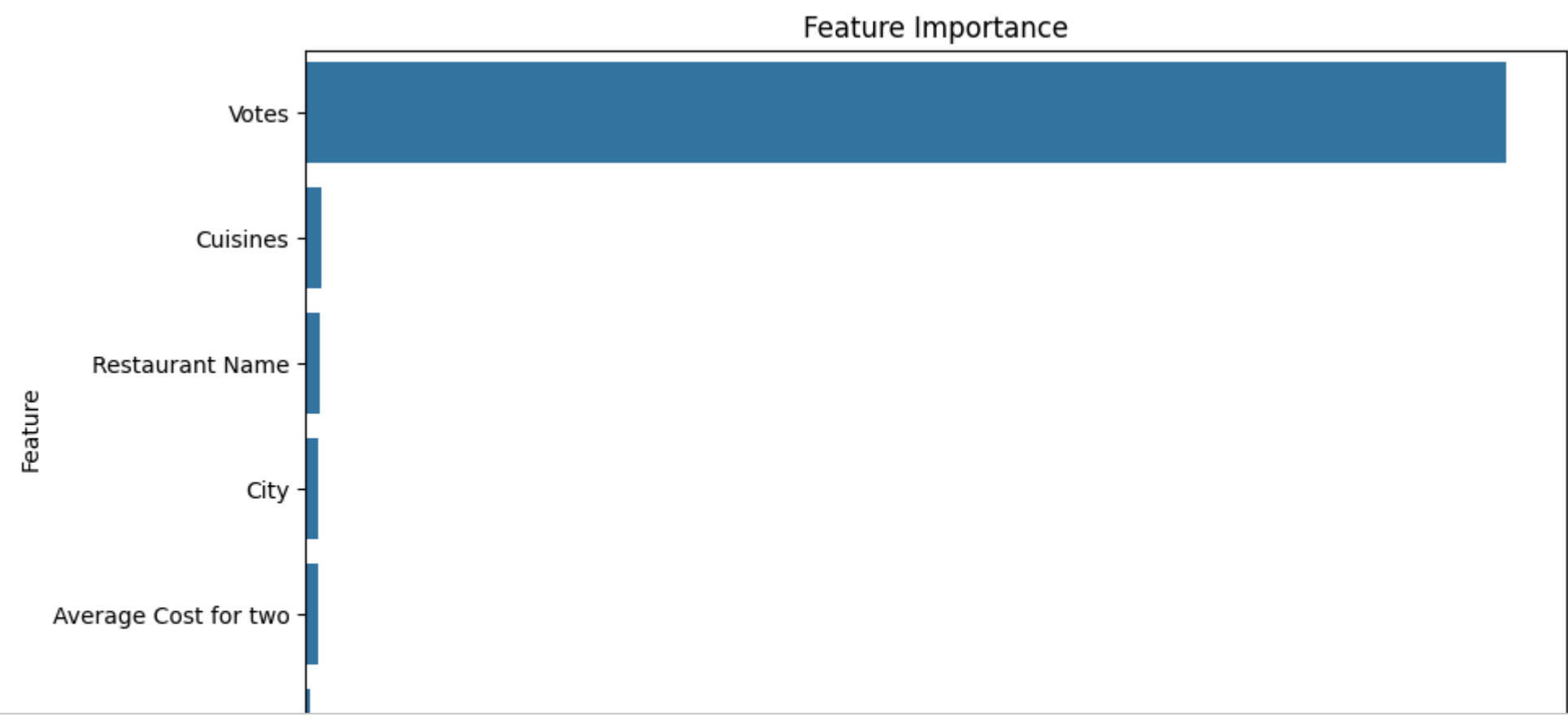
print(importance_df)
```

	Feature	Importance
5	Votes	0.951719
2	Cuisines	0.012785
0	Restaurant Name	0.011586
1	City	0.010687
3	Average Cost for two	0.010144
4	Price range	0.003078

```
plt.figure(figsize=(10,6))

sns.barplot(
    x='Importance',
    y='Feature',
    data=importance_df
)

plt.title("Feature Importance")
plt.show()
```



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Insights

- 1. Votes and price range strongly influence restaurant ratings.
- 2. Certain cuisines tend to receive higher ratings.
- 3. Restaurant location also impacts ratings.
- 4. Random Forest Regressor performed effectively for prediction.
- 5. Feature importance analysis helped identify the most influential factors.

Conclusion

This project developed a machine learning model to predict restaurant ratings using restaurant-related features.

The workflow included:

- Data preprocessing